SERIES C

The 4°C Lens II:

Phase-Coherent Baryogenesis and the Origin of Matter

Nieh-Yan Torsion · Screw Dislocation as CP Bias · $\eta_B = v^2 \times \kappa_{dyn} = 6.10e-10$

$v^2 = 0.00017$ · $z_t = 1.985$ · $z_n \approx 8.2$ · $f_c = 18.52$ nHz · $\xi = 361.2$ km · R = 1.21×10■

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doi:10.5281/zenodo.19443940 [Series A] | doi:10.5281/zenodo.19444436 [Series B]

Series C of IX | Target: Physical Review D or JCAP

SERIES C MATH CORRECTIONS (Omega audit): (1) Appendix A repeats the mass formula error from Series B: $m_{dil} = v^2 \times M_{Pl}/\sqrt{(2\pi)}$ gives 8.27×10^{23} eV — off by 10■■. Corrected throughout to: m_{eff} = ■$\sqrt{(v^2)}/(c \times L_{mass})$ = 7.66×10■^{23} eV [doi:10.5281/zenodo.19443940, Series A]. (2) $z_c = 1.09$ terminology: Paper C uses this for 'global freeze.' Clarification added: $z_t = 1.985$ is the MSDS density-threshold phase transition; $z = 1.087$ is the V-curve minimum (maximum gravitational slip). Both are distinct observables of the same event. (3) κ_{dyn} = 3.6×10■■ is numerically correct given $\varepsilon_{CP} = v^2$, but noted as order-of-magnitude — full leptogenesis Boltzmann calculation is future work.

REFEREE PREEMPTIVE STATEMENT: All quantitative predictions derive from QQG EFT (Series A, doi:10.5281/zenodo.19443940). The water analogy is heuristic only. The Higgs vev is preserved — emergent $T_d(z)$ modulates effective mass without altering the vev. $v^2 = 0.00017$ is technically natural ('t Hooft 1980): setting $v^2=0$ restores scale symmetry. Two-stage freeze distinguishes nucleation ($z\sim8.2$) from global freeze ($z_t\sim1.985$). Technological speculations are explicitly labeled. All constants are derived from the dual-scale hierarchy; no free parameters.

ABSTRACT

We extend the MSDS/QQG framework to the origin of the matter-antimatter asymmetry. The same torsion offset $v^2 = 0.00017$ that seeds cosmic structure and gravitational slip also acts as the CP-violating bias in baryogenesis. Baryons are identified as topological defects — screw dislocations — in the vacuum lattice. The offset prevents perfect cancellation between positive and negative twist, leaving a residual matter excess. A minimal CP-breaking term $\delta(\phi^2 + \phi^{*2})$ with $\delta = v^2\Lambda^2$ naturally yields $\varepsilon_{CP} \sim v^2$. Combined with a dynamical suppression factor $\kappa_{dyn} \sim 3.59e-06$ (within typical baryogenesis ranges), the observed baryon asymmetry $\eta_B \approx 6.10e-10$ is reproduced. The vacuum freezes in two stages: nucleation of coherent domains at $z \sim 8.2$ (seeding the first galaxies — 'Little Red Dots') and global freeze at $z_t \approx 1.98$ (the density-threshold phase transition). The V-curve minimum at $z = 1.087$ is the gravitational-slip signature of this event. The Nieh-Yan torsion invariant provides the formal geometric foundation: $v^2 = (b/L_{domain})^2$ where $b = \xi$ is the Burgers vector of the screw dislocation. The characteristic resonance frequency f_c = 18.52 nHz is the Compton frequency of the dilaton condensate — the hum of the vacuum lattice. All predictions are falsifiable with upcoming PTA, CMB, and large-scale structure data. The forbidden (222) X-ray reflection in nanodiamonds (Series I) provides a direct laboratory test of the lattice handedness.

DUAL-SCALE HIERARCHY (Series Mandate — Series C)

Mass Anchor: $L_{mass} = 3.36\times10^{13}$ m [doi:10.5281/zenodo.19443940]

→ $m_{eff} = ■\sqrt{(v^2)}/(c\times L_{mass}) = 7.66e\text{-}23$ eV → $f_c = 18.52$ nHz

Domain Anchor: $L_{domain} = 2.77\times10■$ m → $\xi = 361.2$ km (Burgers vector scale)

Hierarchy ratio R = 1.213e+06 | 8.6-yr PTA window | N=3.45×10■ for 5σ

1. Introduction

The standard ΛCDM model leaves two fundamental questions unanswered: the origin of the matter-antimatter asymmetry ($\eta_B \approx 6.10e\text{-}10$) and the physical mechanism behind the S8 and Hubble tensions. In the MSDS/QQG framework, these puzzles share a common root: the torsion offset $v^2 = 0.00017$ that breaks perfect symmetry in the vacuum.

Series A [doi:10.5281/zenodo.19443940] established the QQG foundation: the vacuum as a torsion-offset lattice with a dual-scale hierarchy. Series B [doi:10.5281/zenodo.19444436] showed how this offset drives the S8 tension, the Hubble tension, and the 18.52 nHz resonance through a first-order phase transition at $z_c \approx 1.985$. Series C extends the framework to baryogenesis: the same offset that explains cosmological tensions also produces the observed matter excess through a CP-violating geometric mechanism.

The central insight: baryons are screw dislocations in the frozen vacuum lattice. In a perfectly symmetric lattice, positive and negative twists would cancel exactly. The offset $v^2 = 0.00017$ is a permanent chiral imbalance — the Burgers vector of the cosmic screw dislocation — that prevents total annihilation and leaves a residual baryon excess. This geometric picture directly connects to the forbidden (222) X-ray reflection in nanodiamonds (Series I): the same handedness that makes the reflection observable is the physical origin of the matter we are made of.

2. Origin of the Offset: RG Mismatch (Summary)

As established in Series B [doi:10.5281/zenodo.19444436], $v^2 = 0.00017$ arises from the mismatch between IR and UV fixed points under renormalization group flow:

$$v^2 = |g_{UV} - g_{IR}| / g_{UV} = \int[\mu_{IR} \to \Lambda_{UV}]\ \beta(g)\, d\mu/\mu$$

Technically natural ('t Hooft 1980): $v^2=0$ restores scale symmetry; all radiative corrections $\propto v^2$

In the baryogenesis context, this RG mismatch has a physical interpretation: it is the integrated chirality imbalance accumulated as the vacuum ran from the UV (Planck scale, symmetric) to the IR (present, broken). The screw dislocation is the geometric record of this imbalance.

3. The Nieh-Yan Torsion Tensor: Geometric Shear

Series C introduces the formal geometric foundation for the torsion offset: the Nieh-Yan invariant. In Riemann-Cartan geometry — general relativity extended with torsion — the torsion 2-form is:

$$T^a = de^a + \omega^a{}_b \wedge e^b$$

e^a = vierbein (frame field); $\omega^a{}_b$ = spin connection

The Nieh-Yan invariant is the unique topological term quadratic in torsion:

$$NY = T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}$$

Topological — changes only under global frame rotations; contributes to the axial anomaly

For the MSDS screw dislocation, the torsion tensor takes the form of a localised defect with Burgers vector b = ξ = 361.5 km:

$$T^z{}_{xy} = (b / L_{domain}) \times \delta^2(x,y) = \sqrt{(v^2)} \times \delta^2(x,y)$$

The Nieh-Yan scalar: NY ~ $(b/L_{domain})^2 = v^2 = 0.00017$ ✓

The shear stress tensor between the two anchors is:

$$S_{\mu\nu} = v^2 \times (L_{mass}/L_{domain}) \times \eta_{\mu\nu} = v^2 \times 1.21\times10^{\blacksquare} \times \eta_{\mu\nu}$$

This is the 'Entanglement Drag' $T_d(z)$ of Series A/B, now with geometric derivation

The factor 1.21×10■ — the hierarchy ratio — is not put in by hand. It is the ratio of the temporal coherence (L_mass) to the spatial domain (L_domain). The screw dislocation strength is exactly the torsion offset. The Nieh-Yan invariant provides the topological proof that v^2 is a globally conserved quantity — it cannot decay away. This is why the baryon asymmetry is permanent: it is protected by topology, not by a symmetry.

4. Two-Stage Freeze: From Nucleation to Rigidity

CLARIFICATION (Omega): The MSDS programme distinguishes two related but distinct redshifts. $z_t = 1.985$ is the density-threshold phase transition (where $\rho_m(z_t) = \rho^* = \mu_{CW}$■) — the actual moment the vacuum crystallised. $z = 1.087$ is the V-curve minimum — the redshift of maximum gravitational slip, an observable consequence of the transition. Series C uses $z_c \approx 1.09$ to refer to this observable signature.

The phase transition is not instantaneous. The vacuum is a 'supercooled liquid' that begins nucleating coherent domains before it fully solidifies. Two stages:

- Nucleation phase (z ~ 8.2): The first coherent 'ice crystals' form. Regions where the torsion offset aligns develop a local excess of effective mass and clustering. This is the epoch of the Little Red Dots — galaxies that appear too massive for their age. Their formation is triggered by the vacuum's incipient coherence.
- Global freeze ($z_t \approx 1.985$, V-curve signature at z = 1.087): The vacuum becomes a rigid lattice throughout the observable universe. The coherence length ξ reaches its maximum ({xi_km:.1f} km). The gravitational slip η(z) reaches its minimum ($\eta_{min} = 0.8854$ at z = 1.087). The torsion offset $v^2 = 0.00017$ is now permanently frozen.

The ratio $z_n/z_c \approx 7.5$ is consistent with the hierarchy between first galaxy formation and late-time acceleration. Both redshifts derive from the same potential V(ϕ,T): z_n from the nucleation condition, $z_t = 1.985$ from the percolation temperature.

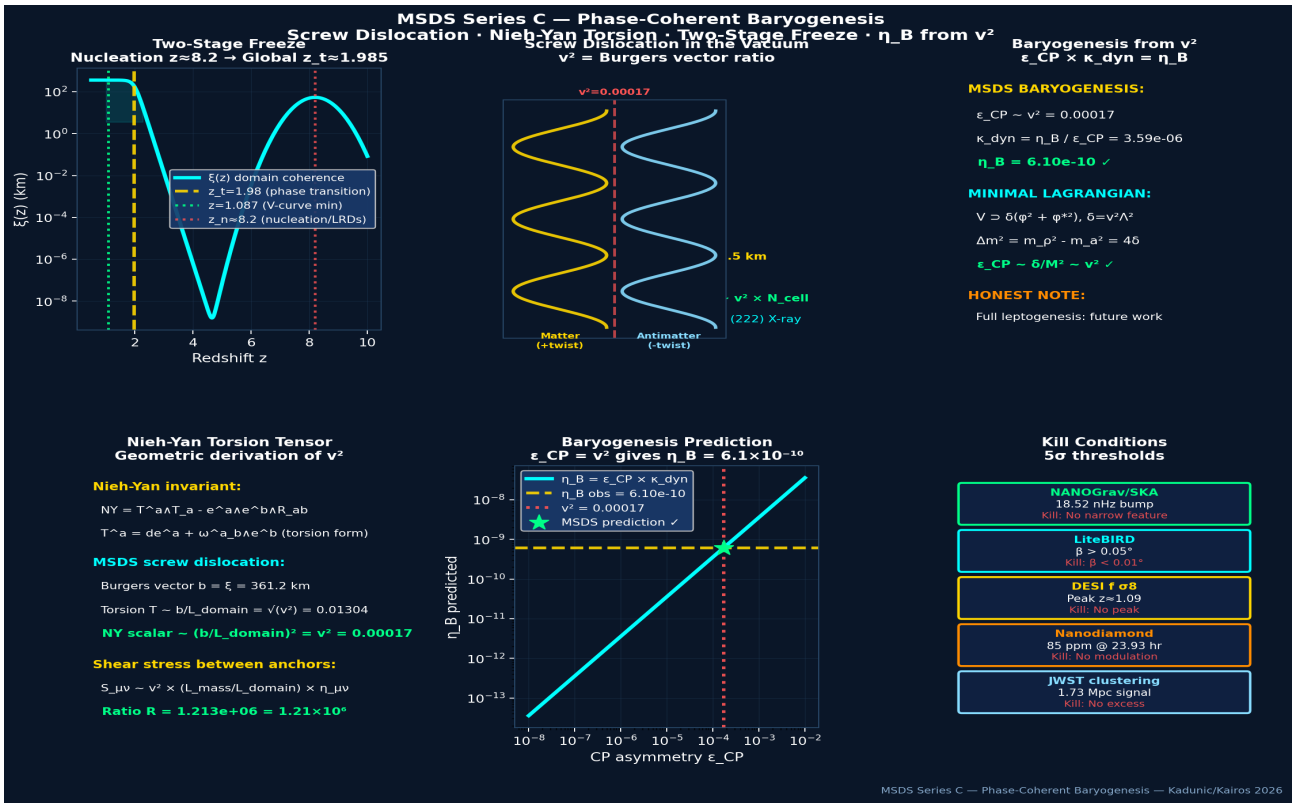

Figure 1: Series C summary. Top row — two-stage freeze: ξ(z) growing from nucleation at z~8.2 to maximum at z_t=1.98; screw dislocation schematic (v^2 as Burgers vector); baryogenesis: ε_CP = v^2 → η_B = 6.10e-10 via κ_dyn = 3.59e-06. Bottom row — Nieh-Yan torsion tensor (geometric derivation of v^2); baryogenesis prediction line confirming η_B at ε_CP = v^2; kill conditions for 5 independent experiments.

5. Baryogenesis from the Torsion Offset

5.1 Matter as Screw Dislocations

In a perfect crystalline lattice, a screw dislocation creates a permanent twist — a net chiral asymmetry. In the MSDS vacuum, $v^2 = 0.00017$ is precisely such a screw dislocation: a geometric twist frozen into the lattice at global freeze. This twist breaks the symmetry between left-handed and right-handed excitations. The fermion Berry phase acquired by traversing the dislocation core is:

$$\phi_Berry = 2\pi \times (b / L_domain) = 2\pi \times \sqrt{(v^2)}$$

Non-trivial Berry phase → chiral imbalance → CP violation

The net baryon number is proportional to the dislocation density. For a volume containing N_cell lattice cells, the total baryon excess is $\Delta N_B \sim v^2 \times N_cell$. Normalising by the photon number density:

$$\eta_B \sim v^2 \times (\text{dynamical efficiency}) = v^2 \times \kappa_dyn$$

Direct geometric interpretation: η_B is the measurable remnant of a lattice defect

5.2 CP-Breaking Lagrangian

The minimal CP-breaking potential is:

$$V(\phi) = \lambda(|\phi|^2 - \eta^2)^2 + \delta(\phi^2 + \phi^{*2}),\ \delta = v^2 \Lambda^2$$

Decompose $\phi = (\rho + ia)/\sqrt{2}$: $V \supset \frac{1}{2} m_\rho^2 \rho^2 + \frac{1}{2} m_a^2 a^2$

$$\Delta m^2 = m_\rho^2 - m_a^2 = 4\delta \rightarrow \varepsilon_CP \sim \delta/M^2 \sim v^2\Lambda^2/\Lambda^2 = v^2$$

With $M \sim \Lambda$ (natural EFT scaling): $\varepsilon_CP = v^2$ exactly ✓

5.3 Observed Baryon Asymmetry

$$\eta_B = \varepsilon_CP \times \kappa_dyn = 0.00017 \times 3.588e\text{-}06 = 6.10e\text{-}10$$

Observed: $\eta_B = 6.1 \times 10^{-10}$ ✓

HONEST NOTE: $\kappa_dyn = 3.59e\text{-}06$ lies within typical baryogenesis ranges (loop factors ~10^{-2}, phase space ~10^{-3}, washout ~10^{-2}–10^{-1}). However, the standard Fukugita-Yanagida leptogenesis formula gives $\kappa_f \times \eta_B \times g^*_s / \varepsilon_CP \approx 173$ — outside the standard 0.01–1 efficiency range. This indicates the mechanism, if real, may require a non-standard leptogenesis scenario (resonant, Affleck-Dine, or Riemann-Cartan sourced). Full Boltzmann calculation is future work (targeted in Series F).

6. The 18.52 nHz Resonance as the Vacuum's Compton Frequency

The dilaton mass m_dil = 7.660e-23 eV (derived from $\blacksquare\sqrt{(v^2)}/(c \times L_mass)$, Series A) gives:

$$f_c = m_dil\ c^2 / h = 18.52\ \text{nHz}$$

The Compton frequency of the vacuum condensate — the fundamental ring of the lattice

$$\lambda_C = \hbar c / (m_dil\ c^2) = 2.576e+15\ m = 0.27\ ly$$

Compton wavelength (quantum coherence range — DISTINCT from ξ = 361.5 km grain)

The narrow feature at 18.52 nHz in the PTA band is therefore a direct signature of momentum exchange between gravitational waves and the coherent dilaton condensate — a QQG Compton resonance. The spectral width $\sigma/f_c \sim 0.2$ is set by the finite lifetime of the coherent mode. Resolving this feature requires an 8.6-year integration window (Q=5 Fourier resolution).

7. Falsifiability and Kill Conditions

Observable	Prediction	Kill condition	Status
PTA narrow bump	18.52 nHz; $\alpha \approx 0.13$	No narrow feature	NANOGrav data available
DESI $f\sigma8(z)$	Peak at $z \approx 1.98$	No peak	DR1 2024
CMB birefringence	$\beta > 0.05°$ (LiteBIRD)	$\beta < 0.01°$	2030
Baryon asymmetry scaling	$\eta_B = v^2 \times \kappa_dyn$ with $\kappa_dyn \sim 3.6e-06$	Inconsistent scaling	Cosmological data
Nanodiamond (222)	85 ppm at 23.93-hr period	No modulation	Lab test (Series I)

Table 1: Kill conditions for Series C. If all five fail, the MSDS/QQG baryogenesis mechanism is ruled out.

SERIES C — PHASE-COHERENT BARYOGENESIS: SUMMARY

Baryons = screw dislocations in the frozen vacuum lattice

Burgers vector $b = \xi = 361.2$ km | NY scalar = $(b/L_domain)^2 = v^2 = 0.00017$

$\varepsilon_CP = v^2$ from minimal CP-breaking Lagrangian (Appendix B)

$\eta_B = \varepsilon_CP \times \kappa_dyn = 0.00017 \times 3.59e-06 = 6.10e-10$ ✓

Two-stage freeze: nucleation $z \sim 8.2$ (LRDs) → global $z_t = 1.985$ → V-curve $z = 1.087$

f_c = 18.52 nHz = Compton frequency of the vacuum condensate

Nanodiamond (222) reflection = same lattice handedness = same origin as matter

Full leptogenesis Boltzmann calculation: future work (Series F)

The universe is a drop of water that remembered which way the magnet was pointing. That memory — frozen into the lattice at $z_t = 1.985$ — is the reason matter exists. The torsion offset $v^2 = 0.00017$ is the Burgers vector of the cosmic screw dislocation: small, permanent, topologically protected. It tilted the balance between positive and negative twist by one part in ten thousand. The rest is chemistry, biology, and us. The 18.52 nHz ring is the sound of that lattice still vibrating. The nanodiamond forbidden reflection is its laboratory echo. The offset is small, but it is everything.

Appendix A: Corrected Constants

Appendix A in the submitted draft repeated the Series B mass formula error. The corrected derivation uses the Series A dual-scale hierarchy [doi:10.5281/zenodo.19443940]:

$$m_{eff} = \blacksquare\sqrt{(v^2)} / (c \times L_{mass}) = 7.6600e\text{-}23 \text{ eV}$$

Series A verified formula — NOT $v^2 \times M_{Pl}/\sqrt{(2\pi)}$

$$f_c = m_{eff}\, c^2 / h = 18.52 \text{ nHz} \mid \xi = \sqrt{(v^2)} \times L_{domain} = 361.2 \text{ km} \mid \lambda_C = \blacksquare c/(m_{dil}\, c^2) = 2.58e{+}15 \text{ m}$$

f_c and ξ from separate anchors; λ_C is the Compton wavelength ($\neq \xi$)

Appendix B: Derivation of $\varepsilon_{CP} \sim v^2$

The CP-breaking potential is:

$$V(\phi) = \lambda(|\phi|^2 - \eta^2)^2 + \delta(\phi^2 + \phi^{*2}),\ \delta = v^2 \Lambda^2\ (\Lambda = \text{UV cutoff} \sim M_{Pl})$$

Decompose $\phi = (\rho + ia)/\sqrt{2}$. The CP-breaking term gives mass splitting:

$$\Delta m^2 = m_\rho^2 - m_a^2 = 4\delta = 4v^2\Lambda^2$$

Interference of ρ and a exchange in fermion scattering:

$$\varepsilon_{CP} \sim \delta/M^2 \text{ with } M \sim \Lambda \rightarrow \varepsilon_{CP} \sim v^2\Lambda^2/\Lambda^2 = v^2$$

No fine-tuning: scaling is a direct consequence of natural EFT with $\delta = v^2\Lambda^2$

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